

Experiment (2)

KINEMATICS OF LINEAR MOTION

2.1 Introduction

Newton's description of Kinematics provides explanations and descriptions of motion of objects regardless of the cause of this motion. Specifically, if we know where a group of objects is (the positions) at a point in time, and also how they are moving at that time (the velocity or momentum), then we can predict where they will be and how they will be moving at any other point in time. The power to do this is extremely useful! In this lab, let us test our ability to make these predictions.

2.2 Objectives

- Investigation of the parameters of motion (distance, velocity and acceleration) in the case of linear motion with uniform acceleration.
- Determination of the gravitational acceleration using a free fall particle.
- Investigation of the parameters of projectile motion of a body.

2.3 Theory

2.3.1. Kinematics of linear motion.

When a particle moves a displacement Δx along x -axis during a time Δt then the average velocity along the x -axis is given by:

$$v_{av,x} = \frac{\Delta x}{\Delta t}$$

In the limit in which the time interval Δt tends to zero, $v_{av,x}$ tends to the instantaneous velocity given by:

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad (2.1)$$

When this velocity changes by Δv_x during a time Δt then the average acceleration along the x -axis is given by:

$$a_{av,x} = \frac{\Delta v_x}{\Delta t}$$

and the instantaneous acceleration is given by:

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} \quad (2.2)$$

When a particle is moving in a straight line with constant acceleration a , with initial velocity v_o and cutting a distance d during a time t , then the one-dimensional kinematic equations for constant acceleration are valid :

$$v = v_o + at \quad (2.3)$$

$$d = v_o t + \frac{1}{2} at^2 \quad (2.4)$$

$$v^2 = v_o^2 + 2ad \quad (2.5)$$

Where v is the velocity at time t .

In this experiment, we verify these relations using two setups; the first is a low friction cart moving on a smooth inclined plane of inclination angle α to the horizontal as shown in Figure 2.1. In this case the acceleration is given by

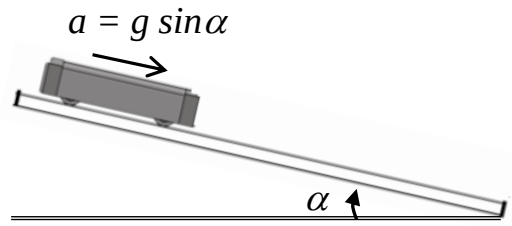


Figure 2.1: Motion along inclined plane.

$$a = g \sin \alpha.$$

In the second setup we determine the gravitational acceleration using a free fall metallic ball. Light barriers are used to get accurately record the location of the ball at specific times.

2.3.2. Kinematics of projectile motion.

A projectile is anybody that is given an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration and air resistance. As shown in Figure 2.2, let the initial velocity vector be given by $\vec{v}_o$ at angle α_0 with x-axis. Then, neglecting the air resistance, the components of position, velocity and acceleration of the projectile motion at any point $p(x,y)$ on the trajectory are given by:

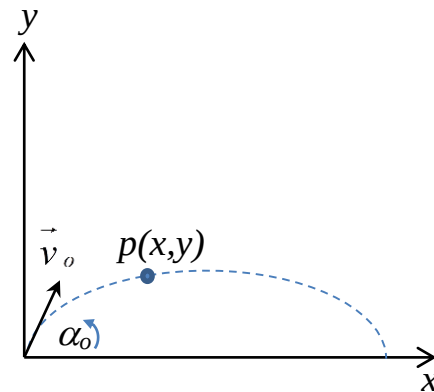


Figure 2.2: The path of projectile.

$$x = (v_o \cos \alpha_o) t \quad (2.6)$$

$$y = (v_o \sin \alpha_o)t - \frac{1}{2}gt^2 \quad (2.7)$$

$$v_x = v_o \cos \alpha_o \quad (2.8)$$

$$v_y = (v_o \sin \alpha_o) - gt \quad (2.9)$$

$$a_x = 0 \quad (2.10)$$

$$a_y = -g \quad (2.11)$$

Equations (2.10) and (2.11) note that we have substituted the acceleration due to gravity at the earth's surface, g , for the acceleration in the y direction and 0 for the acceleration in the x direction. Setting Equation (2.9) to 0 and substituting the time into Equation (2.7) shows that the maximum height of projection h as a function of the angle of projection α_o is given by,

$$h = \frac{v_o^2 \sin^2 \alpha_o}{2g} \quad (2.12)$$

Similarly the maximum range s is given by,

$$x_m = \frac{v_o^2 \sin 2\alpha_o}{g} \quad (2.13)$$

2.4 Task (1): Acceleration of a car moving down an inclined plane , distance- time plot.

2.4.1 Setup

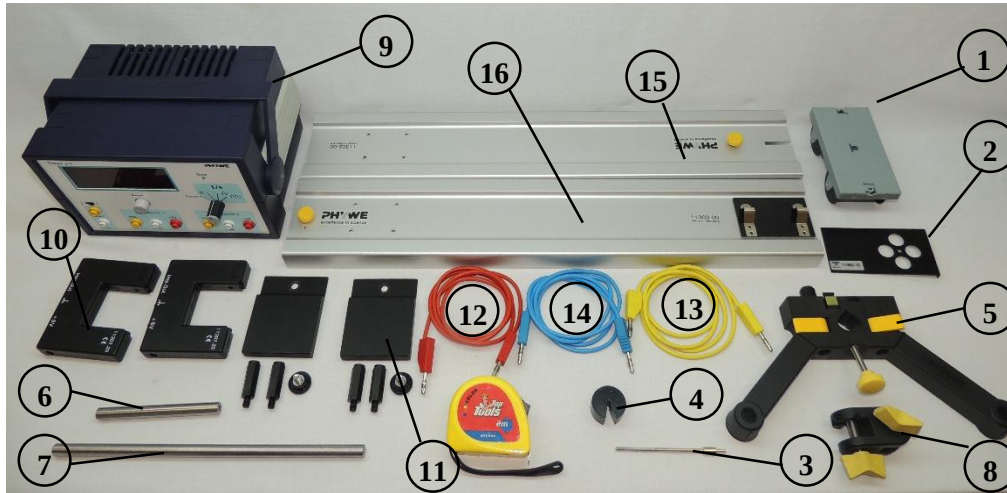


Figure 2.3: Task (1) components.

- | | |
|---------------------------------------|--------------------------------------|
| 1 Measurement/experimental car | 9 Timer 2-1, incl. power supply |
| 2 Shade plate for the measurement car | 10 Light barriers |
| 3 Holding pin | 11 Foot plate for the light barriers |
| 4 Slotted weight, 50 g | 12 Connecting cable, red |
| 5 Support base, variable | 13 Connecting cable, yellow |
| 6 Support rod with hole | 14 Connecting cable, blue |
| 7 Support rod | 15 Track 1, $l = 500 \text{ mm}$ |
| 8 Bosshead | 16 Track 2, $l = 500 \text{ mm}$ |

2.4.2 Procedure

- 1- Connect two tracks as shown in Figure 2.4.
- 2- Connect two spacing bolts and the foot plates onto the light barriers as in Figure 2.5, so that these can stand next to the track, and so that the plate on the car goes through the light barrier.

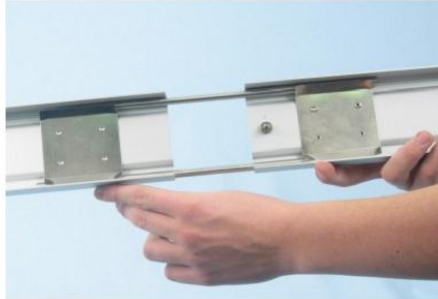


Figure 2.4: Tracks connection.

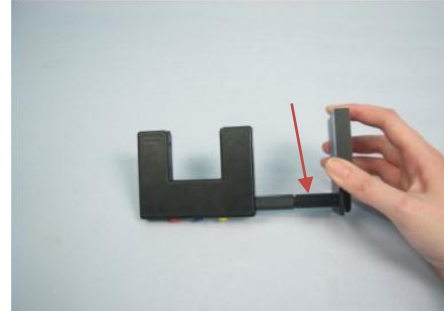


Figure 2.5: Spacing bolts connection.

- 3- Set up the inclination of the track as shown in Figure 2.6 to let the car sliding from the higher end to the lower one.
- 4- Connect the light barriers to the timer as shown in Figure 2.7.



Figure 2.6: Setup the inclination.

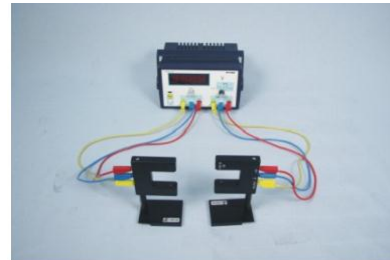


Figure 2.7: Connect the light barriers to the timer.

- 1- Put the light barrier on the () position, the third from the left, to measure the time which the car needs to cover the distance between the light barriers .
- 2- Add the 50 g weight on the car as shown in Figure 2.8 and start the car every time from the same point at the beginning of the track, so that the edge of the car and the edge of the track coincide when seen from the top.



Figure 2.8: Adding the weight.

- 3- Place the first light barrier at a distance of about 10 cm from the end of the track where the car starts, so that the plate on the car interrupts the light barrier's beam immediately after the car starts.
- 4- Place the second light barrier at a distance of $s = 10\text{ cm}$ behind the first as shown in Figure 2.9.

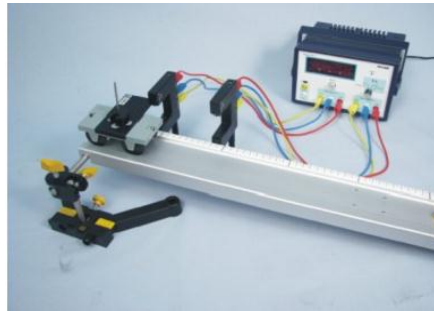


Figure 2.9: Adjusting the light barriers



Don't forget, to press the reset button on the timer, before every measurement.

- 5- Let the car go and catch it after it has gone through the second light barrier.



Make sure to let the car go freely without pushing it, Why?

- 6- Record the time, t , shown by the timer, on your Lab report . This is the time that the car needed to cover the distance $s = 10\text{ cm}$.
- 7- Leave the first light barrier where it is and increase the distance between it and the second light barrier by 10 cm .
- 8- Repeat the steps from 9 to 11 four times by moving the second light barrier 10 cm each time . Record the readings in the report.



Remove the spacing bolts from the light barrier as shown in Figure 2.10, if the plate can no longer run through it .



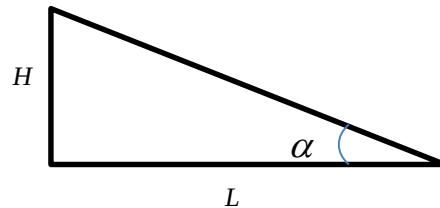
Figure 2.10: Removing spacing bolts.

- 9- Plot the graph with the distance covered, d on the vertical axis and time squared t^2 , on the horizontal axis.
- 10- From the graph calculate the acceleration.
- 11- Compare the result you will get with the theoretical value of the acceleration.

$$a = g \sin \alpha$$

Which α is the inclination angle of the track given by:

$$\alpha = \tan^{-1} \frac{H}{L}$$



2.5 Task (2): Acceleration of a car moving down an inclined plane, distance-speed plot.

2.5.1 Setup

Use the same setup as for the Task (1).

2.5.2 Procedure

- 1- Move the rotary switch of the timer to the " $\sqrt{\quad}$ " position, the second from the left. Now the device will measure the time during which the light beam of the light barrier is cut by the shade plate.
- 2- Remove one light barrier.
- 3- Place the other light barrier at a distance $s = 10 \text{ cm}$ from the edge of cart plate as shown in Figure 2.11.
- 4- Return the cart to the starting position then release it.
- 5- Measure the time Δt which the plate of width $\Delta d = 5 \text{ cm}$ needs to pass through the light barrier.
- 6- Repeat steps 3 to 5 after changing the distance s to the following values $s = 20 \text{ cm}, 30 \text{ cm}, 40 \text{ cm}$ and 50 cm .
- 7- Plot the graph between the squared value of the instantaneous speed v_s on the vertical axis and the distance on the horizontal axis.
- 8- From the graph calculate the acceleration.
- 9- Compare the result you get with the theoretical value

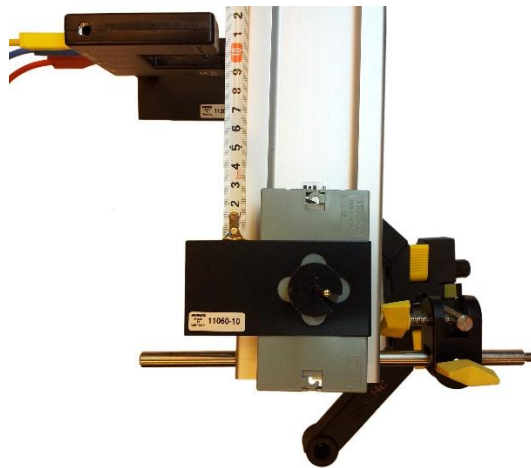


Figure 2.11: The light barrier at a distance 10 cm from the edge of cart plate .

$$a = g \sin \alpha$$

2.6 Task (3): Determine the Gravitational acceleration using a free fall.

2.6.1 Setup

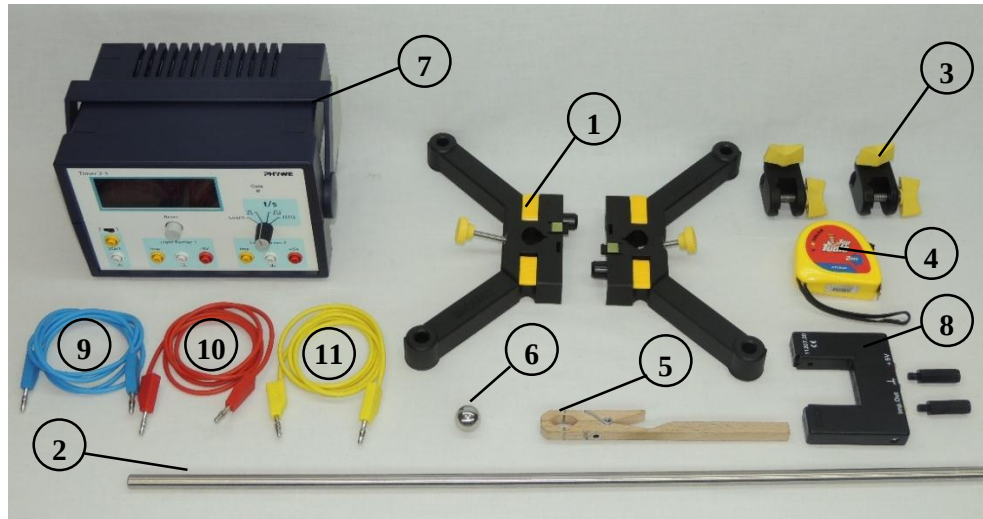


Figure 2.12: Task (3) components.

- | | |
|--------------------------------------|---------------------------------|
| 1 Support base | 7 Timer 2-1, incl. power supply |
| 2 Support rod | 8 Light barrier |
| 3 Bosshead | 9 Connecting cable, blue |
| 4 Measuring tape | 10 Connecting cable, red |
| 5 Steel-ball releasing clamp | 11 Connecting cable, yellow |
| 6 Steel ball, $d = 19,05 \text{ mm}$ | |

2.6.2 Procedure

- 1- Connect the support shaft to the light barrier, so it can be held in the bosshead as shown in Figure 2.13.
- 2- Connect the light barrier to the timer as shown in Figure 2.14.



Figure 2.13: Support shaft connection.

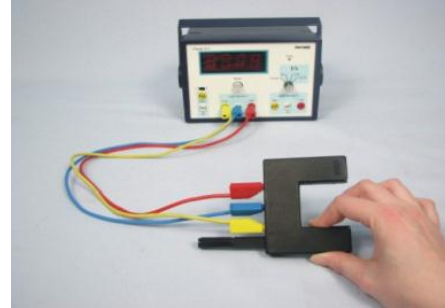


Figure 2.14: Light barrier – timer connection.

- 3- Plug the yellow (11) and blue (9) cables into the sockets of the respective clips. Plug the opposite ends of the cables into the "Start" panel (polarity is not important here) as shown in Figure 2.15.

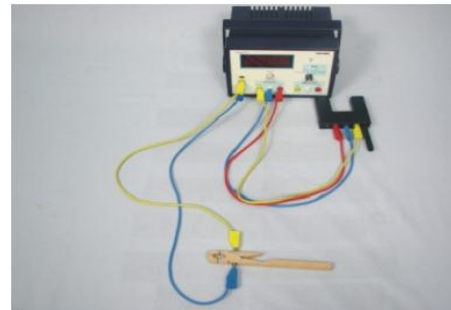


Figure 2.15: Cables connection.

- 4- Set up a support stand as shown in Figure 2.16.



Figure 2.16: Support stand setup.

- 5- Mount the clamp horizontally at the end of the rod as shown in Figure 2.17.
- 6- Mount the light barrier horizontally under the clamp at a distance d , as shown in Figure 2.18. Check that the falling ball will cut the beam of the light barrier when falling.

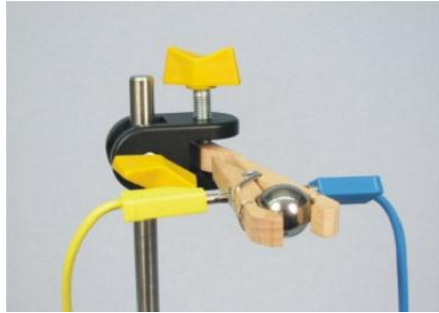


Figure 2.17: Adjusting the clamp.

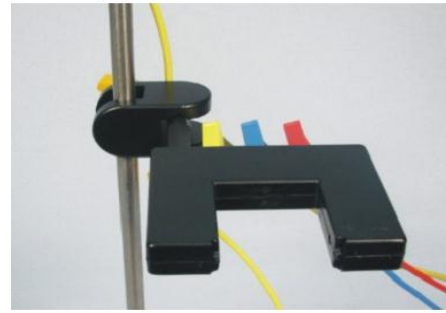


Figure 2.18: Adjusting the light barrier.



Don't **release the ball** before making sure that the ball will not hit the light barrier body. Use the tape or ruler to align the ball to go through the beam and not collide with the device as shown in Figure 2.19.



Figure 2.19: Safety precaution.



Before every measurement **do not** forget to press the reset button on the timer, **after** the ball is placed into the clamp.

- 7- Adjust the timer, in the left position (), so that when the circuit opens at the "Start" input of the timer, the stopwatch starts.
- 8- Adjust the light barrier on the () position, the third from the left. This way, the device shows the time between the interruption of the starting circuit and the interruption of the beam at the light barrier.
- 9- With the measuring tape, set the distance between the lowest point of the ball in the clamp and the beam of the light barrier to $d = 7.5 \text{ cm}$. Always place the ball in the clamp in the same way.
- 10- The light barrier should still be high enough to allow you to catch the ball with your hand under it.



If the ball does not meet the beam at the gate or if it touches the gate's enclosure, or if you obtain times greater than 0.5 s, then repeat the trial.

- 11- Open the clamp as quickly as possible.
- 12- Record the time shown on the timer on Lab report 2.
- 13- Change the distance between the ball and the light barrier's beam to 10, 20, 30, 40 and 50 cm , and repeat the time measurements.
- 14- Plot the graph between the distance d and time t^2 .
- 15- Get the slope of the ($d \text{ vs. } t^2$) plot then use it to calculate the gravitational acceleration.
- 16- Compare the result you will get with the theoretical value 9.81 m/s^2 .



After you finish, please disassemble your setup.